

第四节

多元复合函数的求导法则

一元复合函数 $y = f(u), u = \varphi(x)$

求导法则 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

微分法则 $dy = f'(u) du = f'(u) \varphi'(x) dx$

本节内容:

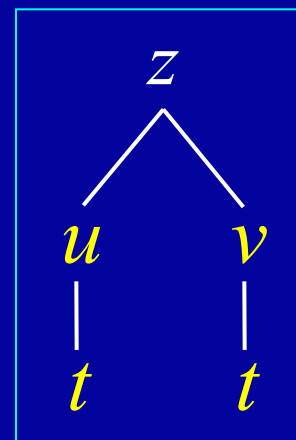
- 一、多元复合函数求导的链式法则
- 二、多元复合函数的全微分



一、多元复合函数求导的链式法则

定理. 若函数 $u = \varphi(t)$, $v = \psi(t)$ 在点 t 可导, $z = f(u, v)$ 在点 (u, v) 处偏导连续, 则复合函数 $z = f(\varphi(t), \psi(t))$ 在点 t 可导, 且有链式法则

$$\frac{dz}{dt} = \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt}$$



证: 设 t 取增量 Δt , 则相应中间变量有增量 $\Delta u, \Delta v$,

$$\Delta z = \frac{\partial z}{\partial u} \Delta u + \frac{\partial z}{\partial v} \Delta v + o(\rho) \quad (\rho = \sqrt{(\Delta u)^2 + (\Delta v)^2})$$



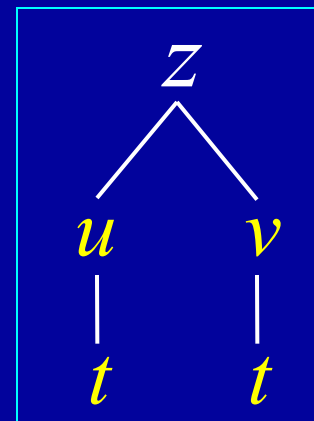
$$\frac{\Delta z}{\Delta t} = \frac{\partial z}{\partial u} \frac{\Delta u}{\Delta t} + \frac{\partial z}{\partial v} \frac{\Delta v}{\Delta t} + \frac{o(\rho)}{\Delta t} \quad (\rho = \sqrt{(\Delta u)^2 + (\Delta v)^2})$$

令 $\Delta t \rightarrow 0$, 则有 $\Delta u \rightarrow 0$, $\Delta v \rightarrow 0$,

$$\frac{\Delta u}{\Delta t} \rightarrow \frac{du}{dt}, \quad \frac{\Delta v}{\Delta t} \rightarrow \frac{dv}{dt}$$

$$\frac{o(\rho)}{\Delta t} = \frac{o(\rho)}{\rho} \sqrt{\left(\frac{\Delta u}{\Delta t}\right)^2 + \left(\frac{\Delta v}{\Delta t}\right)^2} \rightarrow 0$$

($\Delta t < 0$ 时, 根式前加 “-” 号)



$$\frac{dz}{dt} = \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt}$$

(全导数公式)



说明: 若定理中 $f(u, v)$ 在点 (u, v) **偏导数连续** 减弱为 **偏导数存在**, 则定理结论 **不一定成立**.

例如:
$$z = f(u, v) = \begin{cases} \frac{u^2 v}{u^2 + v^2}, & u^2 + v^2 \neq 0 \\ 0, & u^2 + v^2 = 0 \end{cases}$$

$$u = t, \quad v = t$$

易知:
$$\left. \frac{\partial z}{\partial u} \right|_{(0,0)} = f_u(0,0) = 0, \quad \left. \frac{\partial z}{\partial v} \right|_{(0,0)} = f_v(0,0) = 0$$

但复合函数
$$z = f(t, t) = \frac{t}{2}$$

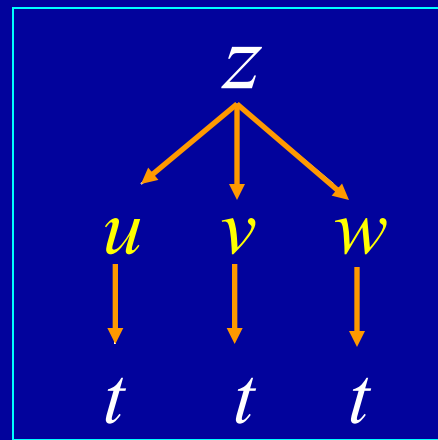
$$\frac{dz}{dt} = \frac{1}{2} \neq \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt} = 0 \cdot 1 + 0 \cdot 1 = 0$$



推广： 设下面所涉及的函数都可微 .

1) 中间变量多于两个的情形. 例如, $z = f(u, v, w)$,
 $u = \varphi(t), v = \psi(t), w = \omega(t)$

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt} + \frac{\partial z}{\partial w} \cdot \frac{dw}{dt} \\ &= f'_1 \varphi' + f'_2 \psi' + f'_3 \omega'\end{aligned}$$

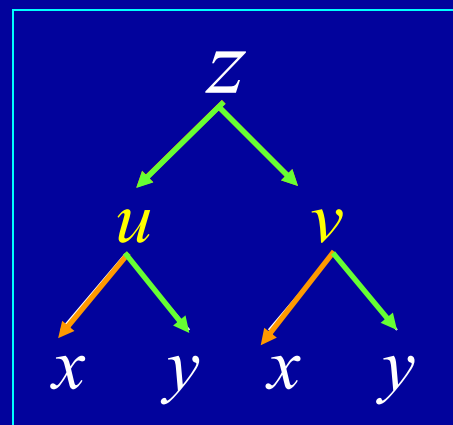


2) 中间变量是多元函数的情形. 例如,

$$z = f(u, v), \quad u = \varphi(x, y), \quad v = \psi(x, y)$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = f'_1 \varphi'_1 + f'_2 \psi'_1$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = f'_1 \varphi'_2 + f'_2 \psi'_2$$



又如, $z = f(x, v)$, $v = \psi(x, y)$

当它们都具有可微条件时, 有

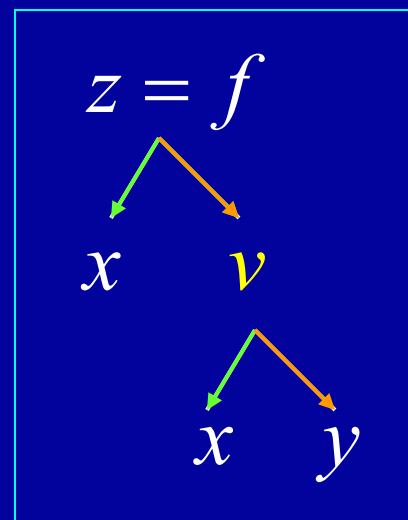
$$\boxed{\frac{\partial z}{\partial x}} = \boxed{\frac{\partial f}{\partial x}} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} = f'_1 + f'_2 \psi'_1$$

$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial y} = f'_2 \psi'_2$$

注意: 这里 $\frac{\partial z}{\partial x}$ 与 $\frac{\partial f}{\partial x}$ 不同,

$\frac{\partial z}{\partial x}$ 表示 $f(x, \psi(x, y))$ 固定 y 对 x 求导

$\frac{\partial f}{\partial x}$ 表示 $f(x, v)$ 固定 v 对 x 求导



口诀:

分段用乘, 分叉用加,
单路全导, 叉路偏导



例1. 设 $z = e^u \sin v$, $u = xy$, $v = x + y$, 求 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$.

解:
$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

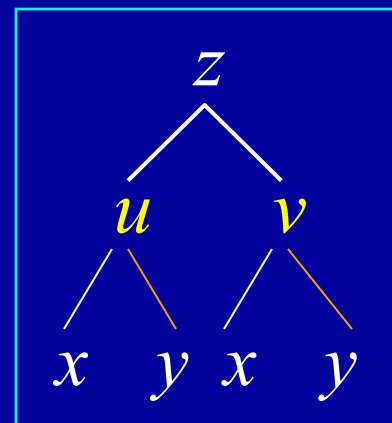
$$= e^u \sin v \cdot y + e^u \cos v \cdot 1$$

$$= e^{xy} [y \cdot \sin(x + y) + \cos(x + y)]$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$= e^u \sin v \cdot x + e^u \cos v \cdot 1$$

$$= e^{xy} [x \cdot \sin(x + y) + \cos(x + y)]$$



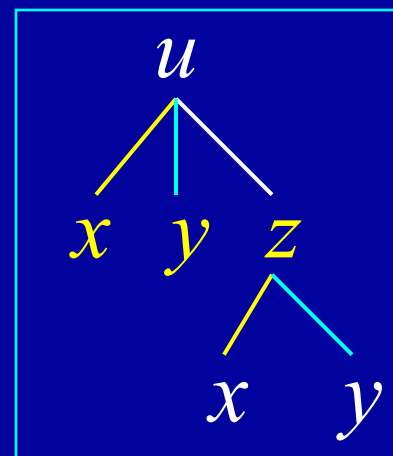
例2. $u = f(x, y, z) = e^{x^2+y^2+z^2}$, $z = x^2 \sin y$, 求 $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$

解: $\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial x}$

$$\begin{aligned} &= 2xe^{x^2+y^2+z^2} + 2ze^{x^2+y^2+z^2} \cdot 2x \sin y \\ &= 2x(1 + 2x^2 \sin^2 y) e^{x^2+y^2+x^4 \sin^2 y} \end{aligned}$$

$$\frac{\partial u}{\partial y} = \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial y}$$

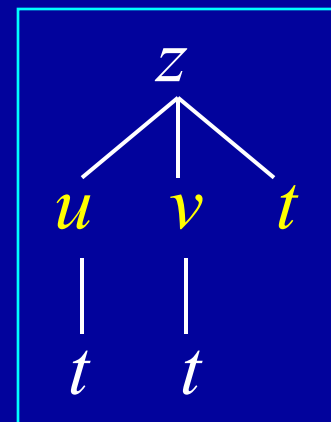
$$\begin{aligned} &= 2ye^{x^2+y^2+z^2} + 2ze^{x^2+y^2+z^2} \cdot x^2 \cos y \\ &= 2(y + x^4 \sin y \cos y) e^{x^2+y^2+x^4 \sin^2 y} \end{aligned}$$



例3. 设 $z = uv + \sin t$, $u = e^t$, $v = \cos t$, 求全导数 $\frac{dz}{dt}$.

解:

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt} + \frac{\partial z}{\partial t} \\ &= v e^t - u \sin t + \cos t \\ &= e^t (\cos t - \sin t) + \cos t\end{aligned}$$



注意: 多元抽象复合函数求导在偏微分方程变形与验证解的问题中经常遇到, 下列两个例题有助于掌握这方面问题的求导技巧与常用导数符号.



例4. 设 $w = f(x + y + z, xyz)$, f 具有二阶连续偏导数,
求 $\frac{\partial w}{\partial x}, \frac{\partial^2 w}{\partial x \partial z}$.

解: 令 $u = x + y + z, v = xyz$, 则

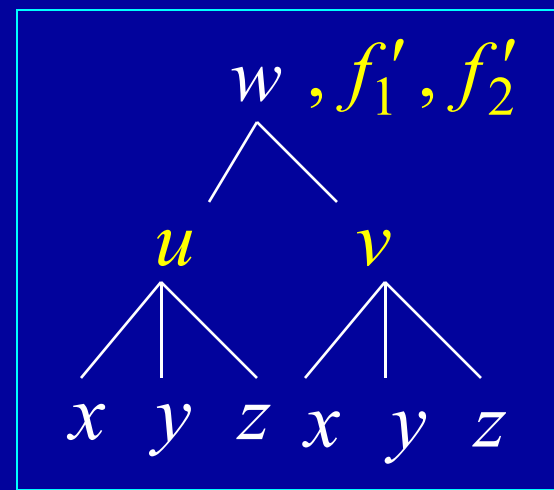
$$w = f(u, v)$$

$$\frac{\partial w}{\partial x} = f'_1 \cdot 1 + f'_2 \cdot yz$$

$$= f'_1(x + y + z, xyz) + \underline{yz f'_2(x + y + z, xyz)}$$

$$\frac{\partial^2 w}{\partial x \partial z} = f''_{11} \cdot 1 + f''_{12} \cdot xy + y f'_2 + yz [f''_{21} \cdot 1 + f''_{22} \cdot xy]$$

$$= f''_{11} + y(x + z)f''_{12} + xy^2 z f''_{22} + y f'_2$$



例5. 设 $u = f(x, y)$ 二阶偏导数连续, 求下列表达式在极坐标系下的形式 (1) $(\frac{\partial u}{\partial x})^2 + (\frac{\partial u}{\partial y})^2$, (2) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$

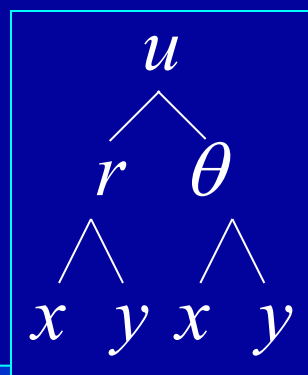
解: 已知 $x = r \cos \theta$, $y = r \sin \theta$, 则

$$r = \sqrt{x^2 + y^2}, \quad \theta = \arctan \frac{y}{x}$$

$$(1) \quad \frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x}$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial \theta}{\partial x} = \frac{\frac{-y}{x^2}}{1 + (\frac{y}{x})^2} = \frac{-y}{x^2 + y^2}$$

$$= \frac{\partial u}{\partial r} \frac{x}{r} - \frac{\partial u}{\partial \theta} \frac{y}{r^2} = \frac{\partial u}{\partial r} \cos \theta - \frac{\partial u}{\partial \theta} \frac{\sin \theta}{r}$$



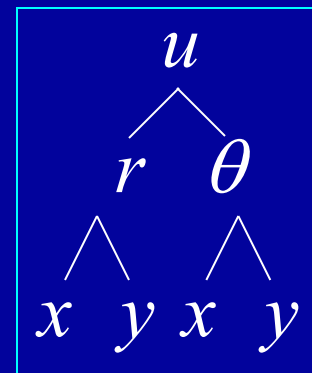
$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial y}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial \theta}{\partial y} = \frac{\frac{1}{x}}{1 + \left(\frac{y}{x}\right)^2} = \frac{x}{x^2 + y^2}$$

$$= \frac{\partial u}{\partial r} \frac{y}{r} + \frac{\partial u}{\partial \theta} \frac{x}{r^2}$$

$$= \frac{\partial u}{\partial r} \sin \theta + \frac{\partial u}{\partial \theta} \frac{\cos \theta}{r}$$

$$\therefore \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta}\right)^2$$



$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cos \theta - \frac{\partial u}{\partial \theta} \frac{\sin \theta}{r}$$



已知 $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cos \theta - \frac{\partial u}{\partial \theta} \frac{\sin \theta}{r}$

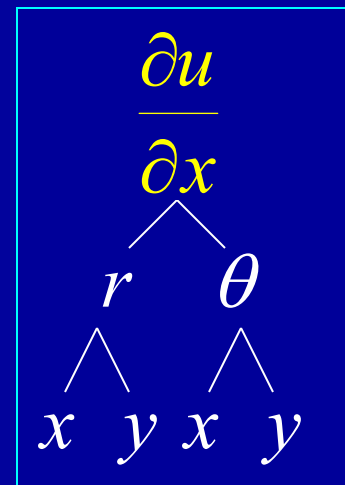
$$(2) \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial r} \left(\frac{\partial u}{\partial x} \right) \cdot \cos \theta - \frac{\partial}{\partial \theta} \left(\frac{\partial u}{\partial x} \right) \frac{\sin \theta}{r}$$

$$= \frac{\partial}{\partial r} \left(\frac{\partial u}{\partial r} \cos \theta - \frac{\partial u}{\partial \theta} \frac{\sin \theta}{r} \right) \cdot \cos \theta$$

$$- \frac{\partial}{\partial \theta} \left(\frac{\partial u}{\partial r} \cos \theta - \frac{\partial u}{\partial \theta} \frac{\sin \theta}{r} \right) \cdot \frac{\sin \theta}{r}$$

$$= \frac{\partial^2 u}{\partial r^2} \cos^2 \theta - 2 \frac{\partial^2 u}{\partial r \partial \theta} \frac{\sin \theta \cos \theta}{r} + \frac{\partial^2 u}{\partial \theta^2} \frac{\sin^2 \theta}{r^2}$$

$$+ \frac{\partial u}{\partial \theta} \frac{2 \sin \theta \cos \theta}{r^2} + \frac{\partial u}{\partial r} \frac{\sin^2 \theta}{r}$$



注意利用
已有公式



$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial r^2} \cos^2 \theta - 2 \frac{\partial^2 u}{\partial r \partial \theta} \frac{\sin \theta \cos \theta}{r} + \frac{\partial^2 u}{\partial \theta^2} \frac{\sin^2 \theta}{r^2}$$

$$+ \frac{\partial u}{\partial \theta} \frac{2 \sin \theta \cos \theta}{r^2} + \frac{\partial u}{\partial r} \frac{\sin^2 \theta}{r}$$

同理可得

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} \sin^2 \theta + 2 \frac{\partial^2 u}{\partial r \partial \theta} \frac{\sin \theta \cos \theta}{r} + \frac{\partial^2 u}{\partial \theta^2} \frac{\cos^2 \theta}{r^2}$$

$$- \frac{\partial u}{\partial \theta} \frac{2 \sin \theta \cos \theta}{r^2} + \frac{\partial u}{\partial r} \frac{\cos^2 \theta}{r}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}$$

$$= \frac{1}{r^2} \left[r \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{\partial^2 u}{\partial \theta^2} \right]$$



二、多元复合函数的全微分

设函数 $z = f(u, v)$, $u = \varphi(x, y)$, $v = \psi(x, y)$ 都可微, 则复合函数 $z = f(\varphi(x, y), \psi(x, y))$ 的全微分为

$$\begin{aligned} dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \\ &= \left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \right) dx + \left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} \right) dy \\ &= \frac{\partial z}{\partial u} \left(\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \right) + \frac{\partial z}{\partial v} \left(\frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \right) \\ &= \frac{\partial z}{\partial u} du + \frac{\partial z}{\partial v} dv \end{aligned}$$

可见无论 u, v 是自变量还是中间变量, 其全微分表达式都一样, 这性质叫做**全微分形式不变性**.



例 6. 利用全微分形式不变性再解例1.

解: $dz = d(e^u \sin v)$

$$= e^u \sin v du + e^u \cos v dv$$

$$= e^{xy} [\sin(x+y) d(xy) + \cos(x+y) d(x+y)]$$

$$= e^{xy} [\sin(x+y)(ydx + xdy) + \cos(x+y)(dx + dy)]$$

$$= e^{xy} [y \sin(x+y) + \cos(x+y)] dx$$

$$+ e^{xy} [x \sin(x+y) + \cos(x+y)] dy$$

所以 $\frac{\partial z}{\partial x} = e^{xy} [y \cdot \sin(x+y) + \cos(x+y)]$

$$\frac{\partial z}{\partial y} = e^{xy} [x \cdot \sin(x+y) + \cos(x+y)]$$



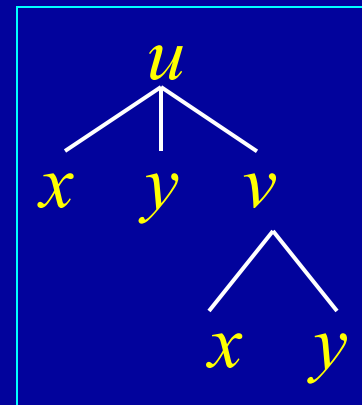
内容小结

1. 复合函数求导的链式法则

“分段用乘，分叉用加，单路全导，叉路偏导”

例如, $u = f(x, y, v)$, $v = \varphi(x, y)$,

$$\frac{\partial u}{\partial x} = f'_1 + f'_3 \cdot \varphi'_1; \quad \frac{\partial u}{\partial y} = f'_2 + f'_3 \cdot \varphi'_2$$



2. 全微分形式不变性

对 $z = f(u, v)$, 不论 u, v 是自变量还是中间变量,

$$dz = f_u(u, v) du + f_v(u, v) dv$$



思考与练习

P81 题7; 8(2); P130 题11

解答提示:

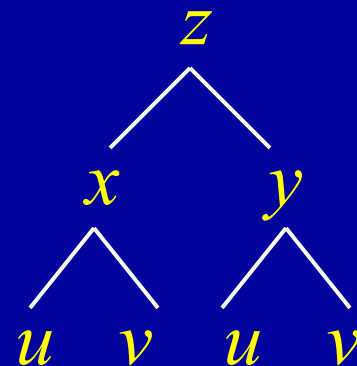
P81 题7 $z = \arctan \frac{x}{y}$, $x = u + v$, $y = u - v$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}$$

$$= \frac{1}{1 + \left(\frac{x}{y}\right)^2} \cdot \frac{1}{y} \cdot 1 + \frac{1}{1 + \left(\frac{x}{y}\right)^2} \cdot \left(-\frac{x}{y^2}\right) \cdot (-1)$$

$$= \frac{y + x}{x^2 + y^2} = \frac{u}{u^2 + v^2}$$

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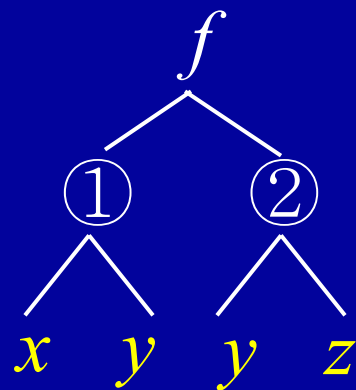


P81 题8(2) $u = f\left(\frac{x}{y}, \frac{y}{z}\right)$

$$\frac{\partial u}{\partial x} = f'_1 \cdot \frac{1}{y} = \frac{1}{y} f'_1$$

$$\frac{\partial u}{\partial y} = f'_1 \cdot \left(-\frac{x}{y^2}\right) + f'_2 \cdot \frac{1}{z} = -\frac{x}{y^2} f'_1 + \frac{1}{z} f'_2$$

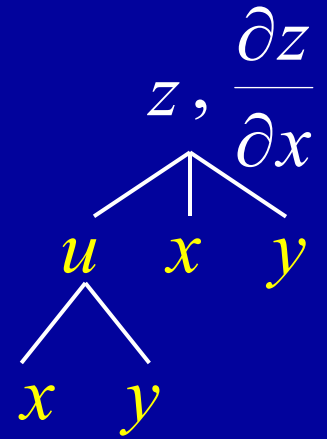
$$\frac{\partial u}{\partial z} = f'_2 \cdot \left(-\frac{y}{z^2}\right) = -\frac{y}{z^2} f'_2$$



P130 题 11 $z = f(u, x, y), u = x e^y$

$$\frac{\partial z}{\partial x} = f'_1 \cdot e^y + \underline{f'_2}$$

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} = & e^y \cdot f'_1 + f''_{11} \cdot x e^{2y} + e^y \cdot f''_{13} \\ & + f''_{21} \cdot x e^y + f''_{23} \end{aligned}$$



作业

P81 2; 4; 6; 9; 10; *12(4); *13



备用题

1. 已知 $f(x, y)|_{y=x^2} = 1$, $f_1'(x, y)|_{y=x^2} = 2x$, 求 $f_2'(x, y)|_{y=x^2}$.

解: 由 $f(x, x^2) = 1$ 两边对 x 求导, 得

$$f_1'(x, x^2) + f_2'(x, x^2) \cdot 2x = 0$$

$$\downarrow f_1'(x, x^2) = 2x$$

$$f_2'(x, x^2) = -1$$



2. 设函数 $z = f(x, y)$ 在点 $(1, 1)$ 处可微, 且

$$f(1, 1) = 1, \quad \left. \frac{\partial f}{\partial x} \right|_{(1,1)} = 2, \quad \left. \frac{\partial f}{\partial y} \right|_{(1,1)} = 3,$$

$$\varphi(x) = f(\underline{x}, \underline{f(x, x)}), \text{ 求 } \left. \frac{d}{dx} \varphi^3(x) \right|_{x=1}. \quad (2001 \text{ 考研})$$

解: 由题设 $\varphi(1) = f(1, f(1, 1)) = f(1, 1) = 1$

$$\begin{aligned} \left. \frac{d}{dx} \varphi^3(x) \right|_{x=1} &= 3 \varphi^2(x) \left. \frac{d\varphi}{dx} \right|_{x=1} \\ &= 3 \left[f_1'(x, f(x, x)) \right. \\ &\quad \left. + f_2'(x, f(x, x)) (\underline{f_1'(x, x) + f_2'(x, x)}) \right] \Big|_{x=1} \\ &= 3 \cdot [2 + 3 \cdot (2 + 3)] = 51 \end{aligned}$$

